



Why is it so Hard to Run a 5G Core on Kubernetes—and What Needs to Change for 6G



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Easy Retro		5G Core on Kubernetes		Prime Directive	
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What Went Well with 5G Core on K8s?		What Needs Improvement for 5G on K8s?			
Starting point for containerization of the core and prep work for 5G		K8s current network abstraction is great for microservices, but not network-heavy stuff that needs low-level controls. Make network concepts first-class citizens in K8s.			
Little, is this K8s native really?		More stateless architecture to avoid service interruptions just because of pod restarts.			
Subnetless		Application modularity (less black box)			
LCM automation (Flowd)		More semantic to underlying infrastructure/platform			
We've made first steps with 5G Ops and BSM		Adapt DNA for networking			
Most NF with exception of AMF, SMF, UPF (because of too much state) are no concern at all		Cloud Native should be in the room when 5G SDRM specs are discussed			
Having different network interface using Muxes		Hardware / Driver integration, think native resource allocation?			
Vendor harmonization thanks to Kubernetes API		Some defaults for observability, logging, metrics, alerts need to follow same standards			
Scaling		Kubernetes VLAN capabilities for traffic going into and out of the cluster			
Five year 5G/6G on some components		Vendors (Ericsson, Nokia etc) just repackaged the previous generation existing solutions. For example Ericsson repackaged PDW to UPF with little to no modification to a containerized resource			
Decoupled microservices		Cloud native config management			
5G Core is usually CNF - it has really accelerating the integration delay towards VNF Model		Stop using dedicated load balancers (L7) and others			
Hands off deployment of entire core stack below 1h		Smaller container resource requirements, lower CPU and memory requirements per container			
Running 4GB binaries as micro services works not so bad		Kick out RAN CP protocols, I believe RAN nodes can also speak http			
Microservices architecture		Replace NETCONF with K8s+GQDs			
Start considering supply chain - still too hand crafted...but progress thanks to OCI		Compatibility matrix hell?			
		OAI dependence is specific, configuration reduces flexibility and ability to multi-vendor platform			
		Declarative and cloud-native configuration mgmt			
		Adapt Ngnix like architecture, Ngnix was a start...			
		Avoid Telco specific format do not reinvent the IT wheels			
		More stateless services			
		More stateless containers			
		Instead of rat you can also use lgo and use lgo to use the connection.			

 Settings

What Needs Improvement for 6G on K8s?



Hands off deployment of entire core sites below 1h

ative resource allocation?

Smaller container resource requirements, lower CPU and memory requirements per container



Got a Question? Please Ask!



Upcoming Events

October 6 2026: Sylva Summit in Prague, Czech Republic

<https://sylvaproject.org/event/sylva-at-open-source-summit-europe/>

November 23-24 2026: Sylva Hackathon in Bern, Switzerland

<https://sylvaproject.org/event/sylva-hackathon/>

November 25-26 2026: CNTF in Bern, Switzerland

<https://oniva.swisscom.com/en/event/cntf-2026>

November 9 2026: FluxCon in Salt Lake City, USA

<https://events.linuxfoundation.org/kubecon-cloudnativecon-north-america/co-located-events/fluxcon/>

March 15 2026: Cloud Native Telco Day in Barcelona, Spain

<https://events.linuxfoundation.org/kubecon-cloudnativecon-europe/>